

Introduction to Bootstrap

Bootstrap is a **CSS framework** used to build responsive websites faster — write less code, use ready-made classes, and ship beautiful UIs in record time.

Why Bootstrap?

- Write Less CSS
- Use Ready Classes
- Build Faster Websites

Real World Usage

Companies use Bootstrap for:

- Landing pages
- Dashboards
- Admin panels
- Responsive websites

Bootstrap at a Glance

Feature	Purpose	Example
Grid System	Responsive layout	12-column grid
Components	Ready UI elements	Navbar, Cards
Utilities	Fast styling	m-3, p-4, text-center
Flex Classes	Alignment	d-flex, justify-content-center

Teaching Tip — Login Example: Show students the difference visually.
Without Bootstrap: padding: 10px 20px; background-color: blue; border-radius: 5px;
With Bootstrap: btn btn-primary

Why Companies Use Bootstrap

Bootstrap dramatically accelerates UI development by providing a complete toolkit of responsive utilities and pre-built components — saving teams hours of custom CSS work.

⚡ Faster Development



📱 Responsive Design

Bootstrap automatically adjusts layout for every device:

- Mobile → Responsive
- Tablet → Responsive
- Laptop → Responsive

🧩 Reusable Components

Ready-made components include:

- Navbar
- Cards
- Buttons
- Alerts

✅ **Teaching Tip:** Explain that companies save significant time using pre-built UI components — no need to reinvent the wheel for every project.



Bootstrap Setup Using CDN



CDN Setup

Add Bootstrap directly inside your HTML — no installation required. Just link the CDN files in your `<head>` and before `</body>`.

Folder Structure

```
project/
├── index.html
├── style.css
└── images/
```

HTML Structure

```
<div class="container">
  <h1>Hello Bootstrap</h1>
</div>
```

CSS CDN

Add in `<head>` tag

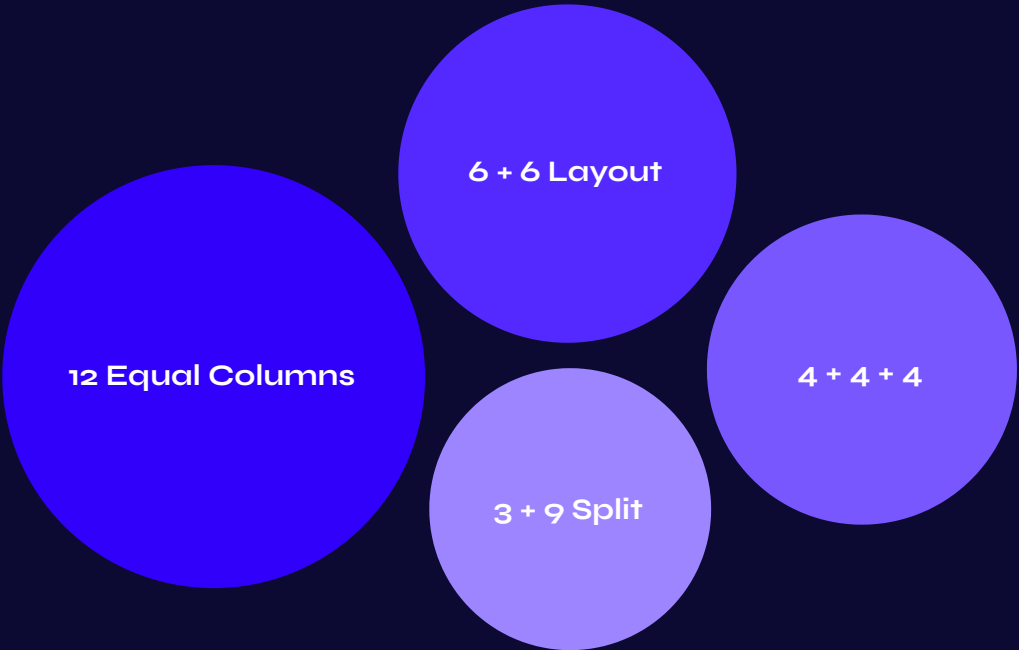
JavaScript CDN

Add before `</body>` tag

① **Teaching Tip:** CDN = Online Bootstrap Link. No installation needed — just copy and paste the link tags into your HTML file and Bootstrap is ready to use instantly.

Bootstrap Grid System

Bootstrap's grid is built on a **12-column layout system** — the foundation of every responsive Bootstrap design. Columns can be combined in any way that adds up to 12.



The grid divides any row into 12 equal parts. Combine them freely to create any layout you need.

Example 1 — Two Columns

Left (col-6)	Right (col-6)
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Example 2 — Three Columns

1 (col-4)	2 (col-4)
3 (col-4)	

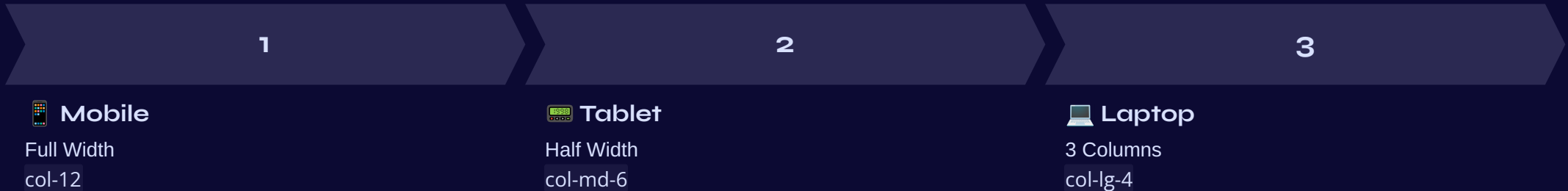
📌 **Teaching Tip:** Use background colors on each column to visually explain how the 12-column grid divides the row. Color-coding makes the concept immediately clear.

Responsive Breakpoints

Bootstrap changes layout automatically based on screen size using **breakpoint classes**. Each breakpoint targets a specific device range.

Breakpoint	Class	Device
Small	sm	Mobile
Medium	md	Tablet
Large	lg	Laptop
Extra Large	xl	Desktop

Visual Explanation



Teaching Tip: Resize the browser window live during the session so students can see the layout shift in real time as breakpoints are triggered.

Spacing and Typography

Margin Classes

m-3

All sides margin

mt-4

Margin top

mb-5

Margin bottom

mx-auto

Center horizontally

Padding Classes

p-3

All sides padding

pt-5

Padding top

px-4

Padding left & right

Typography Headings

Bootstrap provides heading classes `h1` through `h6` for consistent sizing across your project.

Text Alignment

- `text-center` — Center aligned
- `text-start` — Left aligned
- `text-end` — Right aligned

Example Output

Welcome Students

This text is centered using `text-center` and padded using `p-4`.



Teaching Tip — Key Distinction:

Margin = outside spacing (space around the element)

Padding = inside spacing (space within the element)

Bootstrap Colors and Buttons

Color Utilities

text-primary
Blue text color

bg-dark
Dark background

bg-warning
Yellow background

Button Types

Class	Color	Use Case
btn-primary	Blue	Main action
btn-success	Green	Confirm/Save
btn-danger	Red	Delete/Remove
btn-warning	Yellow	Caution
btn-secondary	Gray	Secondary action

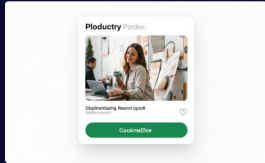
Example Usage

```
<button class="btn btn-primary">Submit</button>  
<button class="btn btn-danger">Delete</button>
```

❏ **Teaching Tip:** Ask students to create all button variations on their own — btn-primary, btn-success, btn-danger, btn-warning, and btn-secondary — as a quick hands-on exercise.

Bootstrap Components

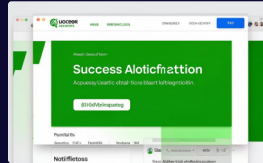
Bootstrap ships with a rich library of ready-to-use UI components that cover the most common interface patterns found in real-world web applications.



Cards

Flexible content containers with image, title, text, and action buttons. Used in product listings and dashboards.

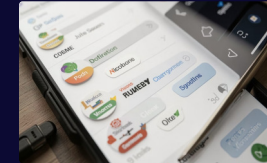
```
<div class="card">
  <h5>Product Card</h5>
</div>
```



Alerts

Contextual feedback messages for user actions. Variants include success, warning, danger, and info.

```
<div class="alert alert-success">
  Login Successful
</div>
```



Badges

Small count and labeling components. Perfect for notification counts, status indicators, and tags.

```
<span class="badge bg-primary">
  New
</span>
```

Real World Usage



Ecommerce Websites



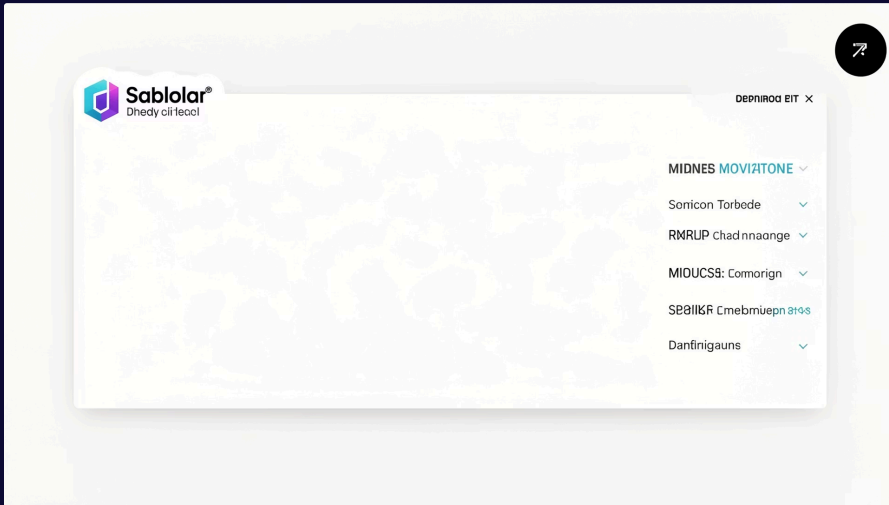
Dashboards



Product Pages

Teaching Tip: Build one complete product card live with students — add an image, title, description, and a button — so they see how components combine in practice.

Flex Utilities



Navbar Example

```
<nav class="d-flex justify-content-between align-items-center">
  <div>Logo</div>
  <div>Menu</div>
</nav>
```

- **Teaching Tip — Key Distinction:**
justify-content = horizontal alignment (left/right/center along the main axis)
align-items = vertical alignment (top/bottom/center along the cross axis)

Enable Flexbox

Start with `d-flex` on the parent container to activate Bootstrap's flexbox utilities.

Justify Content

- `justify-content-between` — Space between items
- `justify-content-center` — Center all items
- `justify-content-start` — Align to start
- `justify-content-end` — Align to end

Align Items

- `align-items-center` — Vertical center
- `align-items-start` — Align to top
- `align-items-end` — Align to bottom

Flex Direction

- `flex-row` — Horizontal layout
- `flex-column` — Vertical layout

Practical Session and Homework

Students will apply everything learned by building a complete responsive webpage using Bootstrap — from navbar to cards layout — entirely from scratch.

Practical Build

Students will build:



Responsive Navbar



Hero Section



Cards Section



Responsive Design

Final Learning Outcomes



Bootstrap Basics



Responsive Design



Components



Grid System



Utility Classes



Flexbox

Homework Assignment

Create a **Product Showcase Page** using Bootstrap only.

Requirements

- Responsive layout
- Bootstrap grid system
- Cards with images and buttons
- Flex utilities for navbar
- Mobile responsive on all devices



Real Project Structure: By completing this session, students will have the confidence and skills to build real-world Bootstrap projects independently — from simple landing pages to full admin dashboards.